

ABSTRACT

Public blockchains continue to struggle with scalability because improving throughput is not as simple as increasing block size or reducing block interval. Larger blocks increase validation and transmission cost, while shorter intervals raise the likelihood of propagation delays, forks, and stale blocks. These limits motivate sharding [2], where transaction processing is divided across multiple parallel shard groups. In this work, we present a configurable SimPy-based discrete-event simulator for evaluating sharded blockchain architectures under controlled workload and network assumptions. The simulator models mining, verification, inter-shard coordination, block dissemination, measured throughput, average block time, and communication overhead. Our goal is not to claim production deployment performance, but to study scaling trends in a controlled environment. Our simulator achieves 1.6M TPS at 256 shards under a local datacenter-like setup and 0.6M TPS in a global WAN setup, showing strong throughput gains from parallel execution. However, the gains are not unbounded: beyond a certain number of shards, coordination traffic, synchronization, and network overhead begin to dominate, leading to diminishing returns.

MOTIVATION

- Bitcoin achieves ~ 7 TPS [1] and Ethereum ~ 15 – 30 TPS — far below the demands of global financial infrastructure
- Increasing block size raises validation and propagation cost; shorter block intervals amplify fork rates and stale blocks
- Production deployments cannot provide the controlled, reproducible experimentation needed to study scaling limits
- Sharding — parallel transaction processing across shard groups [2, 3] — is a leading approach, but its true scaling behavior and overhead crossover points remain poorly understood
- Simulation offers a principled alternative: configurable, noise-free, and reproducible across architectures and network conditions

EXPERIMENTAL SETUP

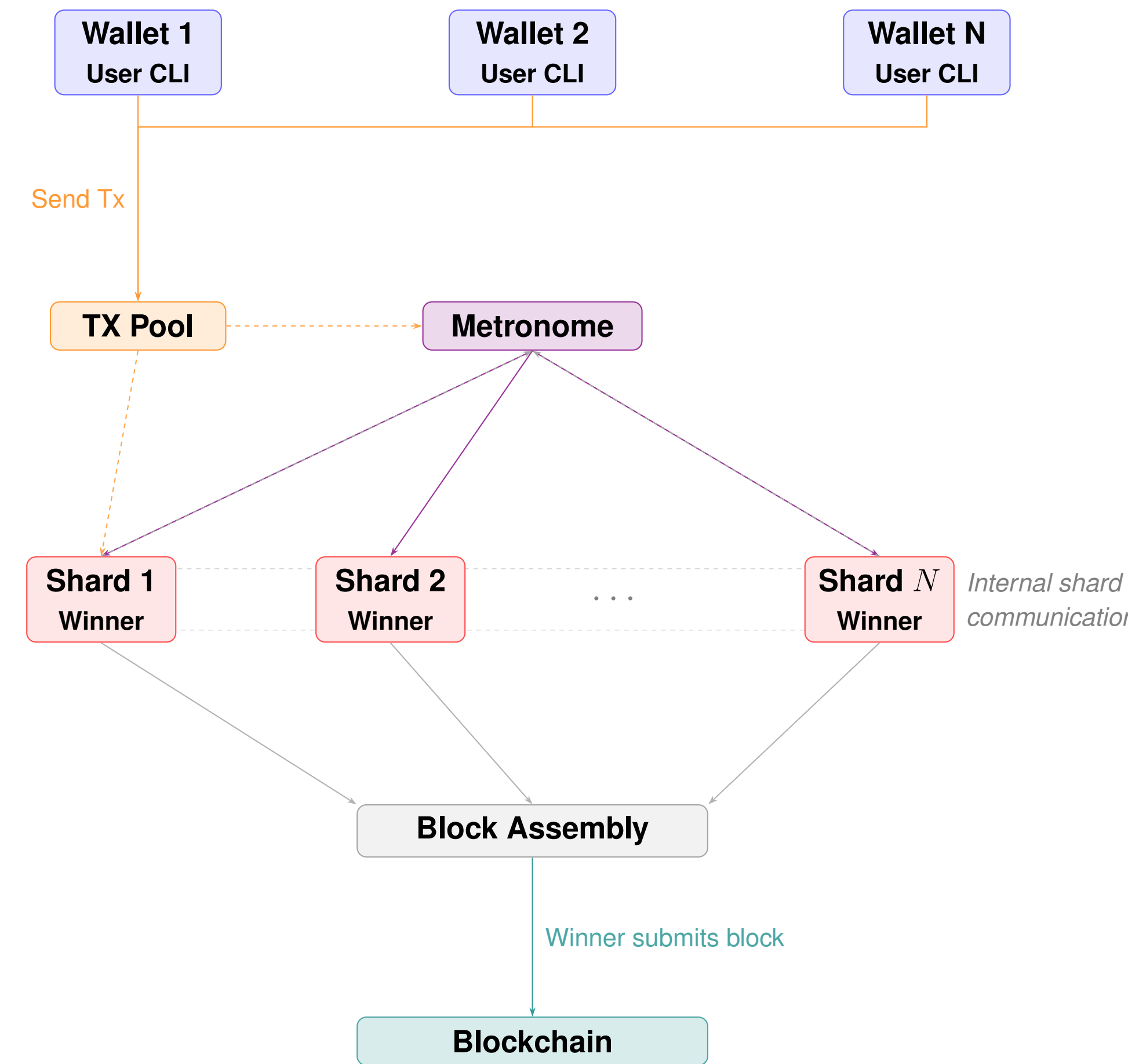
Machine Configuration

Component	Specification
CPU Model	2× Intel Xeon Silver 4108 @ 1.80 GHz
CPU Cores	16 physical / 32 logical (2 sockets × 8 cores)
Turbo Boost	up to 3.00 GHz
L2 Cache	16 MiB total, 16 instances
L3 Cache	22 MiB total, 2 instances
Memory	64 GB RAM
Storage	4× Samsung 970 EVO 250 GB NVMe SSD
OS	Ubuntu 22.04.4 LTS (kernel 5.15.0-171)
Python	3.10.12
SimPy	4.1.1

Simulation Parameters

Parameter	Values
Shard count	1, 2, 4, 8, 16, 32, 64, 128, 256, 512
Block size	1,024 – 524,288 txns per block
Block time	0.001s – 1,200s
Network condition	Local datacenter
Architectures	MEMO-style
Verification cost	Configurable per shard
Bandwidth / RTT	Configurable per network profile

SHARDED ARCHITECTURE DESIGN



- Wallets** submit transactions to the TX Pool; winners claim batches for their shard
- Metronome** broadcasts challenges and collects proofs from all shards each round
- Parallel Shards** compete via PoSpace simultaneously with internal communication
- Block Assembly** merges shard results; winner submits finalized block to the chain

MEMO RESULTS

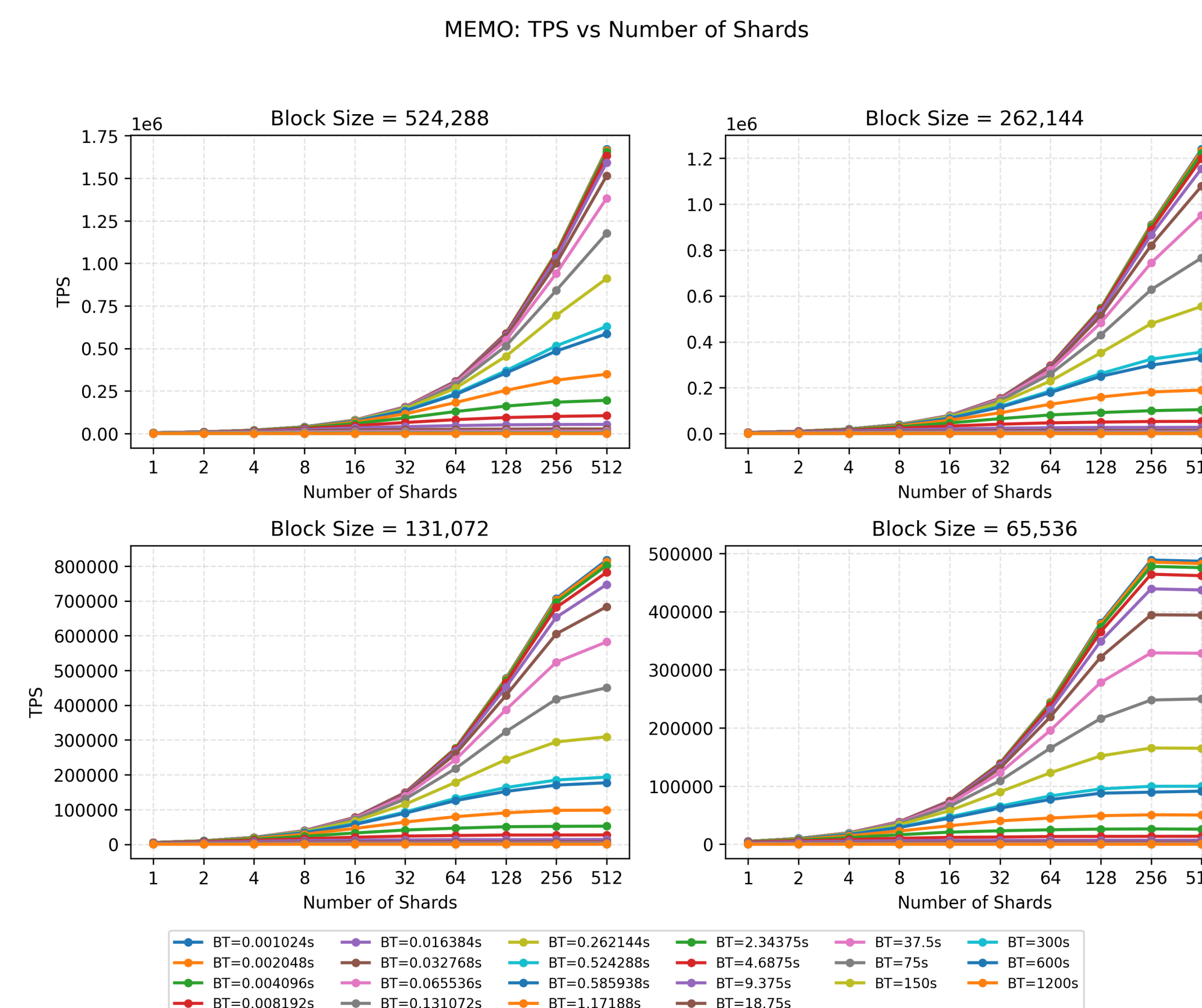


Figure 1: MEMO TPS for large block sizes (65,536–524,288 txns); peaks at 1.4M TPS at 512 shards.

MEMO RESULTS (cont.)

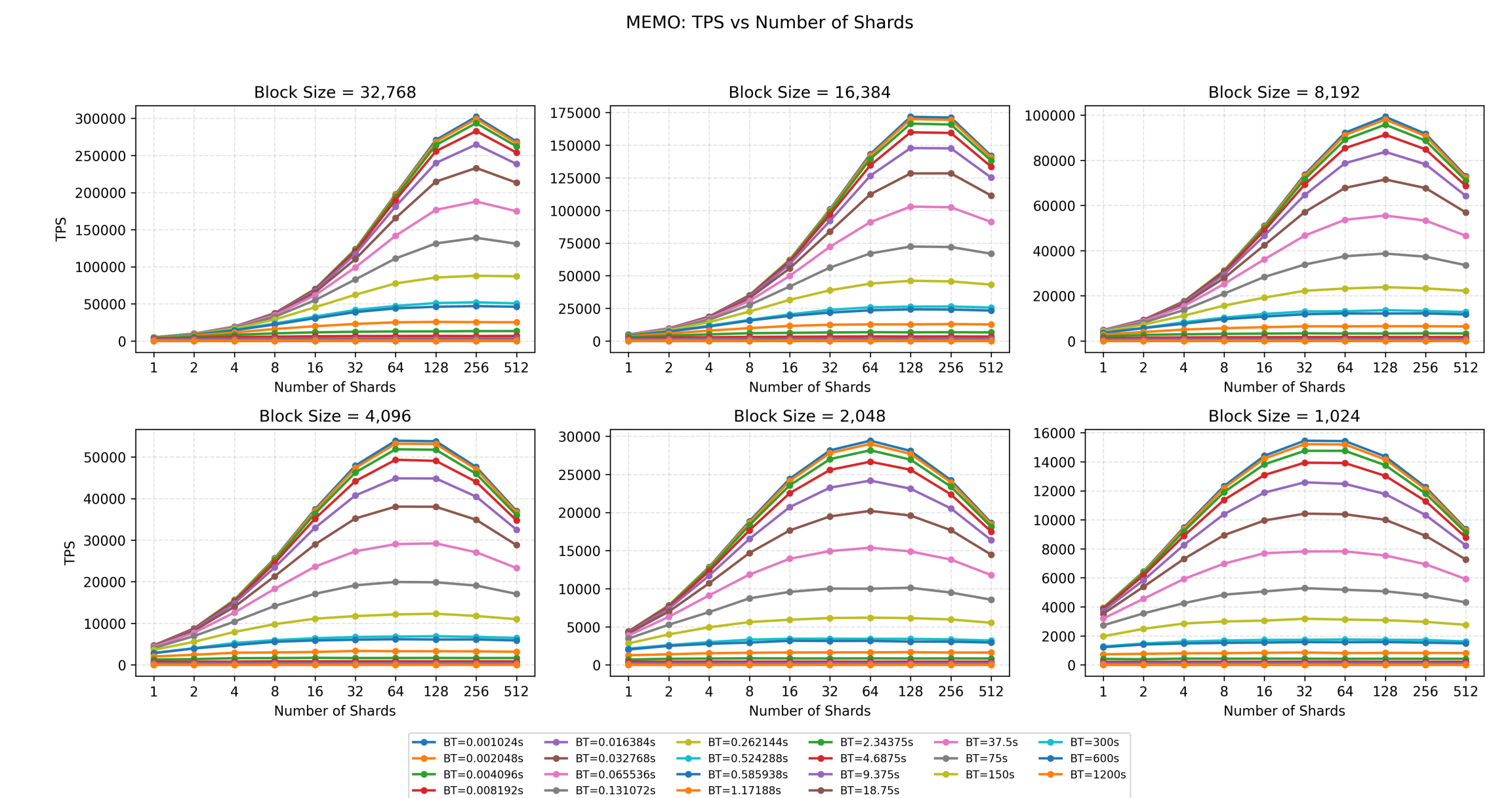


Figure 2: MEMO TPS for small block sizes (1,024–32,768 txns); peaks at 64–128 shards.

- Sharding improves throughput by enabling parallel block production across multiple shards
- Large block sizes with short block times yield the highest TPS, exceeding 1.6M TPS at 512 shards under local datacenter conditions.
- Small block sizes peak between 64–128 shards roughly then exhibit diminishing returns as coordination overhead dominates
- The simulator exposes a clear throughput–latency–overhead tradeoff rather than unlimited linear scaling

CONCLUSION

- Sharding scales throughput via parallelism but gains are bounded by coordination overhead
- Block size and block time are the dominant tuning parameters at scale
- Offers a reproducible simulation basis for sharded blockchain evaluation

FUTURE WORK

- Calibrate simulator using live NEAR/Ethereum testnet measurements
- Validate simulated TPS and finality against live testnet observations
- Extend to atomic cross-shard coordination and fault-tolerance modeling

REFERENCES

- [1] Satoshi Nakamoto. “Bitcoin: A Peer-to-Peer Electronic Cash System”. In: (2008). URL: <https://bitcoin.org/bitcoin.pdf>.
- [2] Alex Skidanov and Illia Polosukhin. *Nightshade: Near Protocol Sharding Design*. 2019. URL: <https://near.org/papers/nightshade>.
- [3] Mahdi Zamani, Mahnush Movahedi, and Mariana Raykova. “RapidChain: Scaling Blockchain via Full Sharding”. In: *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*. 2018, pp. 931–948.